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Economic Impacts of the Recovery and Resilience Facility: New Insights at Sectoral Level and the Case of Germany

Anne Michels, Daria Ciriaci, José Manuel Rueda-Cantuche, Luis Pedauga, Valeria Ferreira, Christina Kattami, Daniel Schulz and Marta Pilati

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Authorised by Luc Tholoniat, Director "Economies of the Member States II".

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Directorate-General for Economic and Financial Affairs

Economic Impacts of the Recovery and Resilience Facility: New Insights at Sectoral Level and the Case of Germany

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Abstract

The Recovery and Resilience Facility (RRF) is the cornerstone of NextGenerationEU, the EU's post-covid Recovery Plan. With a budget of EUR 650 billion, amounting to around 4% of the EU GDP, the RRF finances over 2,800 reform and investment measures aross the EU from 2020 to 2026. To date, most studies on the RRF have focused on the broad macro-economic picture or on specific policy domains and country experiences. This paper introduces a novel approach to enhance the understanding of the RRF. First, by leveraging artificial intelligence, a new sectoral database was created, which classifies all RRF reform and investment measures according to economic sectors and enables more detailed and systematic analysis of macro-economic, sectoral and policy impacts. The paper presents an overview of this classification, across the EU and for each Member State. The sectors benefitting most from the RRF include those related to the green transition, such as energy efficiency, renewable energy production and sustainable transport; those supporting the manufacturing sector and broader business environment, including digitisation, public administration and justice reforms; and those related to societal resilience, such as healthcare, skills and education. We also observe significant synergies between reform and investment measures in those areas. Second, by applying the well-established FIDELIO model to this new sectoral dataset, it is now possible to assess the direct and spillover impacts of RRF-funded investments and to break down these results across 64 economic sectors, for the EU and for each Member State. Looking at Germany by way of illustration, the analysis shows that, assuming full implementation of the 27 plans, the German economy stands to benefit from a medium-term stimulus of about EUR 66 billion (1.6% of Germany's GDP), more than twice the size of Germany's Recovery and Resilience Plan. This reflects both the direct impact of the RRF and significant positive spillovers, driven by the strong integration within the EU Single Market. This paper identifies the sectors that benefit most from these effects and highlights how these novel data and techniques can support further research and evaluations.

JEL Classification: C82, E61, E62, F15, F17, F41, F42, F62.

Keywords: Recovery and Resilience Facility, public investment, structural reforms, economic impact, general equilibium model, FIDELIO, artificial intelligence.

Contact: Anne Michels, anne.michels@ec.europa.eu, Daria Ciriaci, daria.ciriaci@ec.europa.eu, José Manuel Rueda-Cantuche, josem.rcantuche@ec.europa.eu, Luis Pedauga, luis.pedauga@ec.europa.eu, Valeria Ferreira, valeria.ferreira@ec.europa.eu, Christina Kattami, christina.kattami@ec.europa.eu, Daniel Schulz, daniel.schulz@ec.europa.eu and Marta Pilati, marta.pilati@ec.europa.eu. European Commission, Directorate-General for Economic and Financial Affairs and Joint Research Centre.

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ABBREVIATIONS

AIDS	Almost Ideal Demand System
CES	Constant Elasticieties of Substitution
CGE	Computable General Equilibrium
DSGE	Dynamic Stochastic General Equilibirum
EAGLE	Euro Area and Global Economy
ECB	European Central Bank
EV	Electric Vehicle
FIDELIO	Fully Interregional Dynamic Econometric Long-term Input-Output
FIGARO	Full International and Global Accounts for Research in input-Output analysis
GDP	Gross Domestic Product
IO	Input-Output
LES	Linear Expenditure System
NACE	Nomenclature statistique des activités économiques dans la Communauté européenne
NGEU	NextGenerationEU
RRF	Recovery and Resilience Facility
RRP	Recovery and Resilience Plan
SMEs	Small- and medium-sized enterprises

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1. INTRODUCTION

With a budget of EUR 650 billion, amounting to around 4% of the 2023 EU GDP, the Recovery and Resilience Facility (RRF) has played a central role in the EU's economic recovery and resilience post-COVID. While implementation is still progressing rapidly, several studies have begun to discuss its impact (e.g. Pfeiffer et al., 2023, Bańkowski et al., 2024, Miller et al., 2023, Ramboll et al., 2022), and a Mid-Term Evaluation was completed in 2024 to assess progress over its 2020-2026 timeframe.

Studies available so far tend to provide two complementary yet relatively disconnected sets of results. A first strand provides a macro-economic aggregate view, looking at the overall impact of the RRF, based on broad assumptions. A second strand takes a micro-economic perspective, looking at certain projects, measures, policy domains or countries, to illustrate specific impacts.

This paper introduces a novel approach that bridges the gap between aggregate macro-economic assessments and more granular micro-economic analyses, by exploring the impact of the RRF at the sectoral level. This meso-level perspective can enhance our understanding of how RRF-funded measures translate into real economic outcomes across different segments of the economy.

To this end, a new sectoral database was developed using artificial intelligence techniques, building on the RRF's main public reporting platform, FENIX. This database classifies all reform and investment measures undertaken by Member States under their national Recovery and Resilience Plans (RRP) into 241 categories of economic activities. These categories were subsequently grouped into 64 main industries, in line with standard classifications used for statistical and economic analysis in the EU.

This dataset was then used as input for a macro-economic analysis of the impact of RRF-funded investments using the well-established FIDELIO[1] model of the European Commission. This general equilibrium model allows to disentangle the direct and spillover impacts of the RRF, analysed as a demand-driven shock and considering value chain linkages across 64 sectors and bilateral trade between 46 countries.

This work paves the way for a more systematic and detailed analysis of the macro-economic, sectoral and policy impacts of the RRF. This paper presents a first sectoral overview of RRF reform and investment measures according to this economic categorisation, for the EU as a whole and for each Member State. It then provides estimates of the direct and spillover impacts of RRF investments across the EU, assuming the full implementation of the 27 plans and leaving the assessment of the impacts of RRF reforms for future research. Looking at the case of Germany, our findings highlight the significance of direct and spillover impacts in a deeply integrated Single Market, both in aggregate terms and within specific sectors.

The paper is structured as follows. Section 2 reviews the main studies on the economic impacts of the RRF, situating this work within the broader relevant literature. Section 3 introduces the new database developed to classify RRF reform and investment measures and presents a first overview. Section 4 explains how these data were used to provide a quantification of direct and spillover impacts and presents first results. Section 5 discusses the case of Germany by way of illustration. Section 6 concludes on the possible use of this work for further analysis and research.

[1] FIDELIO is a dynamic econometric general equilibrium model with a detailed input-output modelling of European industries. FIDELIO was developed by the Joint Research Centre (JRC) and is run under a partnership between the JRC and DG GROW within the European Commission. As part of this collaboration, which began in 2024, FIDELIO is being used to conduct case studies on the effects of industry-relevant policies, as well as to provide ad hoc policy support. FIDELIO is being upgraded to include an SME dimension and extend its capabilities in relation to priorities of DG GROW. Rocchi, P., Reynès, F., Hu, J., Pedauga, L., Cai, M., Boonman, H., Rueda Cantuche, J.M. (2025) FIDELIO 4 manual: Equations and data sources, JRC Technical Report, Publications Office of the EU, Luxembourg.

2. ANALYSIS OF ECONOMIC IMPACTS OF THE RRF TO DATE

The scale of the RRF, its new features, its recent and fast-paced implementation, and the breadth of its activities, which vary according to Member States, make it inherently challenging to study and assess. Such difficulties are compounded by the fact that the RRF has evolved over time – with significant revisions of plans and take-up of loans in 2023 following the war of aggression against Ukraine – and that a substantial share of its more than 2,800 reform and investment measures are still being rolled out, with new developments almost on a daily basis (see box below).

Box 1: **The Recovery and Resilience Facility**

The RRF is the centrepiece of NextGenerationEU (NGEU), the EU's plan to recover from the COVID-19 pandemic crisis, enhance the resilience of the Union and drive EU's policy priorities. At least 37% of the envelope of each Member State's Recovery and Resilience Plan (RRP) is allocated to green objectives and 20% to digital objectives. The plans include both reform and investment measures, which were agreed in light of Country-Specific Recommendations adopted at EU level. Progress is assessed based on the achievement of defined milestones and targets (7,131 in total): one key novelty of the RRF is that it functions as a performance-based instrument, with regular disbursements based on proven results. This also gives transparency and predictability for actors at all levels.

The Facility entered into force in February 2021, providing financing for measures that have taken place since February 2020, which was the start of the pandemic. The measures supported by the Facility need to be completed by 31 August 2026. Each Member State's financial envelope of grants was calculated based on population, inverse GDP per capita and relative unemployment rate, with 30% of it readjusted in 2022 to account for real GDP loss in 2020 and 2021. Member States could also request loan support, of up to 6.8% of their 2019 GNI, until August 2023 (see Figure 1 for RRF envelopes by Member State).

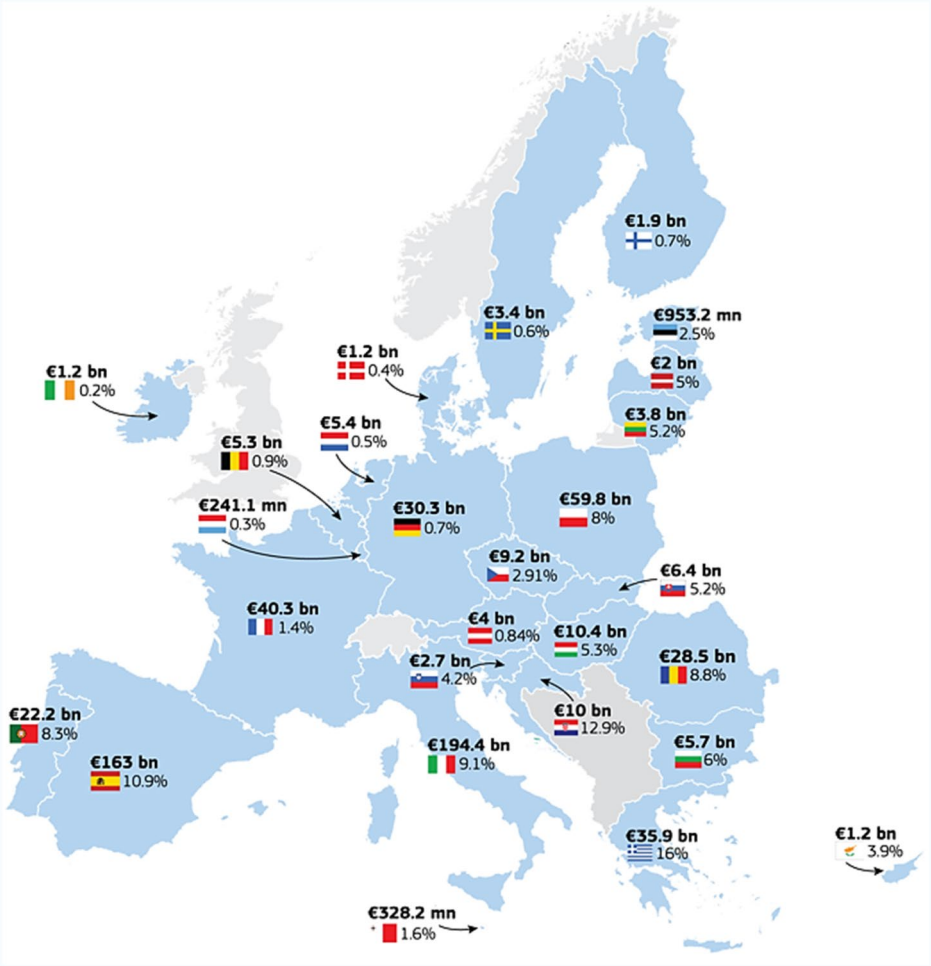
Following Russia's war of aggression against Ukraine, the EU launched REPowerEU in May 2022 to diversify energy supply, save energy and step up clean energy production. REPowerEU mobilised EUR 19 billion of additional grants through the sale of Emission Trading System allowances and EUR 2 billion of funds transferred from the Brexit Adjustment Reserve. 26 Member States revised their RRPs to introduce additional REPowerEU chapters.

After these adjustments and new loan uptakes, the RRF envelope increased in late 2023 from EUR 500 billion to now EUR 650 billion, of which EUR 359 billion are grants and EUR 291 billion are loans. As of 13 May 2025, disbursements under the RRF stand at EUR 311 billion (48% of the envelope). 30% of milestones and targets have been assessed as fulfilled and an additional 22% reported as already completed by Member States.

A lot of information is available publicly,[2] yet, it can be challenging for observers and researchers to extract, structure and analyse the data in a way that enables sophisticated economic assessments.

[2] The main website is: https://commission.europa.eu/business-economy-euro/economic-recovery/recovery-and-resilience-facility_en. Each national Recovery and Resilience Plan provides a detailed description of measures and corresponding milestones and targets for their roll-out and completion. The European Commission assessed these plans and monitors their delivery. The Council of the European Union endorsed them and agrees to regular disbursements. A scoreboard of indicators was put in place at the request of the European Parliament (see: Recovery and Resilience Scoreboard). Progress with the RRF is reported in Annual Reports and regularly discussed within the European Parliament and with the Member States.

Figure 1: **RRF envelopes by Member State (in EUR billion and in % of 2023 GDP)**



Source: European Commission services.

A comprehensive analysis of the RRF would need to capture the impact of its novel framework, which combines both reform and investment measures. This involves assessing not only their individual impact but also their interaction, since many reform measures aim to improve the framework conditions for investments. In practice, however, assessing the impact of reforms is challenging, as their effects are difficult to isolate and typically take time to be captured in economic indicators. Estimating the impact of investments prior to their full implementation is also complex, but it is more feasible and can be approached through ex-ante estimations.

As a result, economic studies of the RRF so far have typically produced two complementary but rather disconnected sets of findings. A first strand adopts a macro-economic aggregate perspective, assessing the impact of the RRF based on broad assumptions essentially linked to the overall stimulus effect, without capturing the more granular impacts of reform or investment measures. A second strand takes a more micro-economic and case-specific approach, focusing on selected projects, measures, policy areas or countries to illustrate specific impacts, without providing a comprehensive and systematic analytical framework. This paper seeks to bridge the gap between these two approaches by allowing the analysis of impacts at sectoral and industry level. This meso-level approach enables a more nuanced understanding of how RRF measures affect different parts of the economy, combining the breadth of macro-economic modelling with the specificity of sectoral analysis.

2.1 OVERVIEW OF MAIN OVERALL AND SPECIFIC STUDIES

Watt and Watzka (2020) conducted one of the earliest assessments of the RRF's potential macro-economic impact, by using the NiGEM macro-econometric policy model. Their analysis showed substantial increases in public capital stocks across the EU, suggesting that the RRF could play a significant role in mitigating output losses from the COVID-19 pandemic crisis and in reducing economic divergence within the EU. They found an average GDP gain in the order of 0.3% per year (Watt and Watzka, 2020).

Subsequently, Pfeiffer et al. (2023) applied the European Commission's QUEST Dynamic Stochastic General Equilibrium (DSGE) model to quantify the NGEU's spillover impacts across EU Member States. Their results suggested that spillovers would contribute to approximately one-third of the average GDP impact, with some highly integrated economies deriving most of their GDP gains from such spillovers. This work also suggested that coordinated investments under the RRF would amplify growth effects beyond individual recipient countries.

Building on this work and using updated data, the European Commission's Mid-Term Evaluation of the RRF (European Commission, 2024) found that NGEU could increase the EU's real GDP every year, reaching a peak increase of 1.4% in 2026 and remaining above 0.6% annually until 2032. Approximatively one-fourth to one-third of this impact is attributed to spillover impacts across Member States. Employment is also projected to see a short-run increase of 0.8%. The growth impact, measured as the peak percentage increase in GDP, is expected to vary across Member States, with Greece (4.5%), Croatia (4%) and Spain (3.5%) estimated to benefit the most relative to the size of their economy. Spillover impacts are projected to be particularly pronounced for small open Member States with smaller RRPs.

The European Central Bank (ECB) conducted several studies on the economic impact of the RRF, using its EAGLE (Euro Area and Global Economy) DSGE model alongside complementary semi-structural modelling approaches. They analyse the RRF impacts through three main channels: fiscal stimulus, structural reforms, and reductions in risk premiums (Bańkowski et al., 2022). In their most recent study (Bańkowski et al., 2024), they find that, under a scenario of successful implementation, NGEU could increase euro area GDP by up to 0.9% by 2026 and 1.2% by 2031.

In parallel to the growing body of research on the macro-economic impact of RRF investments, increasing attention is being devoted to the impact of structural reforms, the assessment of which remains one of the key challenges in evaluating the overall effectiveness of the Facility. This challenge is not unique to the RRF. Garcìa et al. (2024) reviewed existing tools to analyse the economic impact of structural reforms and concluded that both econometric approaches and macro-economic models have limitations in producing a robust quantitative ex-ante assessment of reforms.

Pfeiffer et al. (2023) employed a model-based benchmarking approach to assess the potential macro-economic gains from structural reforms across EU Member States. They simulated the effects of closing half of the gap between national performances and that of EU's best performers, thereby reflecting ambitious reforms for countries with significant distance to the frontier. In addition, the Commission's Directorate-General for Economic and Financial Affairs (2016) quantified the medium- to long-term macro-economic effects of selected structural reforms in Italy, France, Spain, and Portugal. This approach involved translating individual reform measures into changes in the structural indicators of the QUEST model, thereby allowing for an estimation of their macro-economic impacts.

Based on a qualitative approach, the European Commission's Mid-Term Evaluation of the RRF (European Commission, 2024) finds that NGEU has played a crucial role in incentivising Member States to implement key reforms, including many of macro-economic significance that had long been recommended under the European Semester of economic policy coordination. This assessment is supported by Bańkowski et al. (2024), who highlight how the RRF has accelerated the implementation of structural reforms aligned with EU-level Country-Specific Recommendations. Their analysis suggests that, in some cases, the long-term impact of reforms could exceed that of investments. The impact of selected reforms has also been explored in specific cases, such as Italy's judicial system reform (D'Andrea et al., 2023) and Spain's labour market reform (Anghel et al., 2023).

Finally, there is a growing body of research on the micro-economic impact of the RRF with studies examining specific projects, country-level impacts and governance dynamics shaped by the Facility's performance-based design. For instance, several studies have focused on the climate and energy-related impacts of RRF measures (e.g. Hindriks et al., 2022; Ramboll et al., 2022; Miller et al., 2023), while others have assessed the contribution to the digital transition (Demertzis et al., 2024; Empoli et al., 2023). Additionally, governance-related research has provided valuable insights into the strengths and limitations of the RRF implementation framework, with Zeitlin et al. (2024) evaluating its governance model and Begg et al. (2024) analysing the implications of performance-based budgeting.

The European Parliament's Research Service (2025) recently published a European Impact Assessment of the RRF, including a literature review of research papers on key parameters of the RRF.[3] These studies offer multiple valuable insights into the RRF implementation and effectiveness.

2.2 A NEW APPROACH TO ENRICH AND CONNECT MACRO- AND MICRO-ANALYSIS

Two important lessons can be drawn from the existing and emerging literature on the impact of the RRF. First, a more systematic study requires structured, detailed and comprehensive data, together with a modelling framework capable of processing such granular information. Second, any comprehensive assessment must factor in not only the direct impacts of individual plans, but also the spillover impacts: even discounting for the expected positive direct and spillover impacts of reforms, the simultaneous implementation of public investment measures across 27 EU Member States within a deeply integrated Single Market generates substantial cross-border effects.

Against this background, and given the advancing implementation of the RRF, it has become important to bridge the gap between aggregated macro-economic analyses and targeted, project-specific assessments, to offer a more comprehensive and systematic overview of potential impacts.

As a follow-up to the European Commission's Mid-Term Evaluation of the RRF, different additional techniques have been tested, but they all showed some limitations.[4] However, with the progress of the RRF's implementation, the availability of new information in FENIX and the increasing performance of AI tools, it is becoming possible to create new databases in order to document and analyse the RRF in a more systematic and detailed manner.

[3] European Parliament Research Service, (2025), *Recovery and Resilience Facility: State of play and way forward - European Implementation Assessment,* available here Recovery and Resilience Facility: State of play and way forward - European Implementation Assessment | Think Tank | European Parliament. In addition, the European Court of Auditors has produced Special Reports on different features and policy domains of the RRF.

[4] In the lead up to this project, several alternative avenues were explored such as using information regarding the "top 100" final beneficiaries, data held by national administrations and insights from the European Public Tender Database. These options were eventually set aside due to limited data operationability.

3. A NEW DATABASE FOR MACRO-ECONOMIC, SECTORAL AND POLICY ANALYSIS

In recent months, a new sectoral database was established to categorise the 2,878 investment and reform RRF measures[5] into 241 industries, following the standard NACE classification of Eurostat.[6] This classification was carried out using the European Commission's internal AI tools, which leveraged textual descriptions of the RRF measures, as detailed in the annexes of the Council Implementing Decisions, to categorise the measures by economic sector. To ensure accuracy, the classification underwent three rounds of robustness checks by Commission staff.

The database covers the entire RRF over its full implementation period 2020-2026. Reform and investment measures are classified separately. It includes data on actual yearly disbursements over the period 2020-2024, as well as projected yearly disbursements for 2025-2026, disaggregated at both country and measure levels. The cut-off date for the incorporation of amendments to the plans was 15 January 2025, but the database can be updated in the future to reflect subsequent revisions.

3.1 OVERVIEW OF RRF REFORMS BY KEY POLICY DOMAINS

We have classified the 1,131 RRF reform measures based on the sector targeted by each reform. For example, if a reform measure aims to reduce CO2 emissions by providing incentives for the purchase of electric vehicles, it will be classified under 'manufacture of motor vehicles' (NACE code 29.1).[7]

Such categorisation builds on the NACE code classification and follows an economic sector perspective. It thus differs from other categorisations under the RRF framework such as the use of intervention fields for climate and digital tagging, and the pillar reporting system that assigns each measure to policy areas.[8]

The reform-related data highlights the broad scope of the RRF (see Figure 2 below). Energy market reforms (126), education reforms (105), the modernisation of the public administration (91) and justice, anti-corruption and anti-fraud (84) are among the most prominent areas of intervention.

For illustrative purposes, we have regouped these areas into five generic headings: green; business; institutions; health and social; and skills and labour. We express the results in percentage of the total number of RRF reforms.

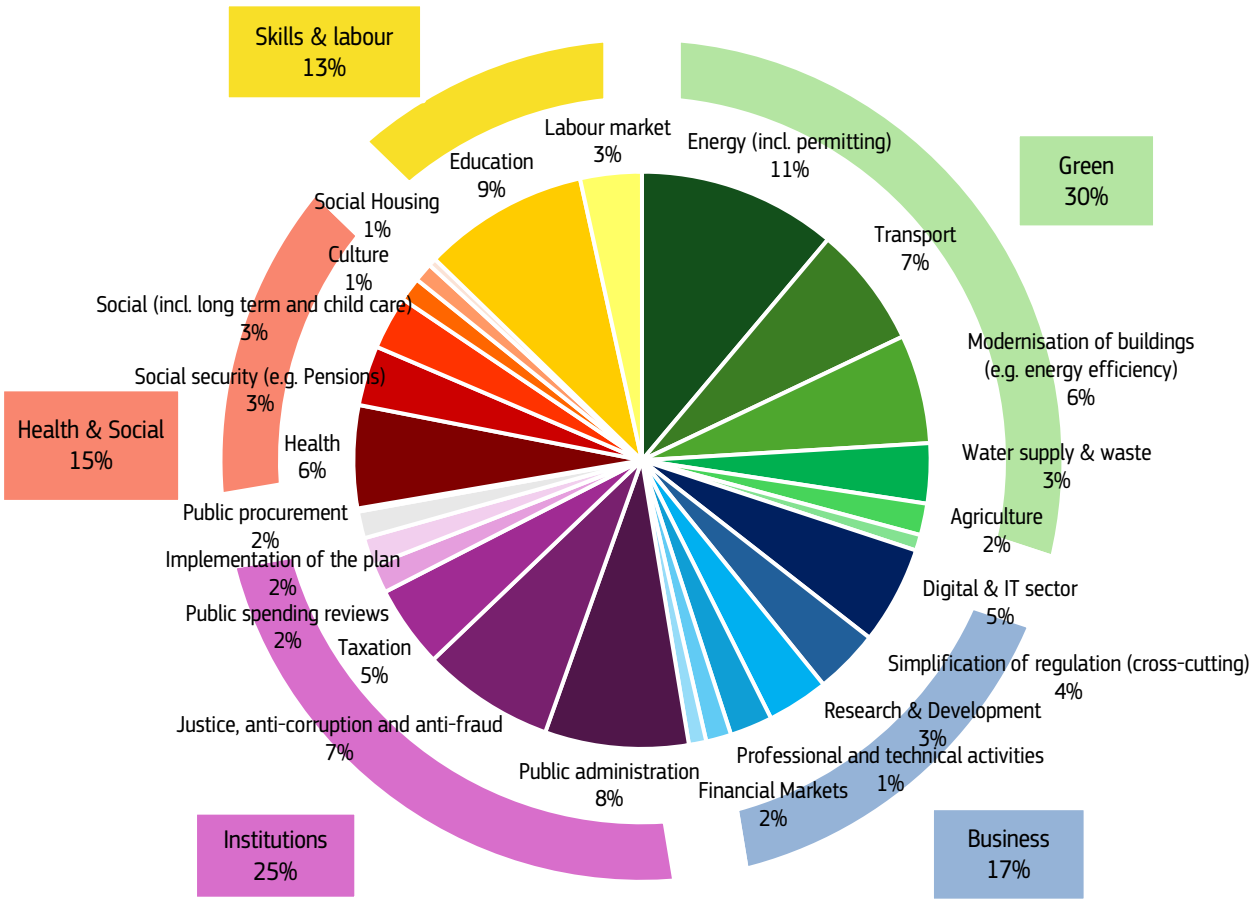
[5] Out of the 2,878 measures, 1,131 are 'reforms' and 1,747 are 'investments'. To bring even more precision to the analysis, we classified them at sub-measure level, which is also the level at which the costing of the RRPs is done. By our cut-off date of mid-January 2025, the Council Implementing Decisions endorsing the 27 plans included 2,878 measures; 4,687 including sub-measures. A sub-measure typically refers to a specific investment or reform component, such as a concrete project, programme, or legislative action, that contributes to the overall objective of the broader measure. For example, one measure in the Austrian plan referring to green investment by enterprises has 6 sub-measures that address different aspects of the green transition, such as energy-saving investments, zero emission mobility and the renovation of buildings.

[6] See NACE Rev. 2 - Statistical classification of economic activities - Products Manuals and Guidelines - Eurostat.

[7] In cases where a measure is linked to multiple sectors, a dual classification system was applied (primary and secondary NACE codes). see Annex I for more details on the classification.

[8] The intervention field classification serves to define a measure's climate or digital contribution under the RRF. This is different from our analysis here: for instance, looking at the 'water supply and waste category' (NACE code 35), one may find measures which are classified as green and/or digital under the RRF intervention field system. The pillar reporting system follows yet another – policy – logic: it attributes measures to six main policy areas and it is used to monitor progress under the Recovery and Resilience Scoreboard mentioned above.

Figure 2: **Overall distribution of reforms by key policy domains, in % of total number of reforms (total = 1,131)**

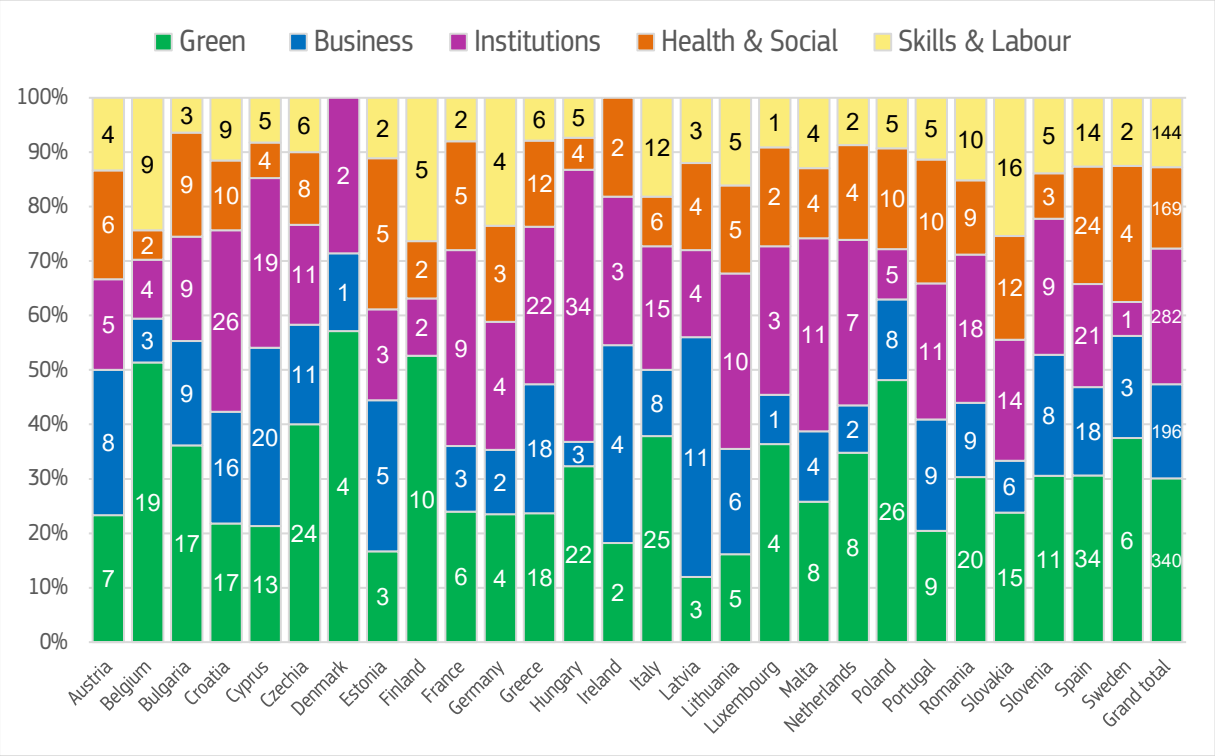


Source: European Commission services.[9]

The policy mix of RRF reforms varies across Member States (see Figure 3 below). Considering only the five generic headings, all Member States dedicate a significant share of reforms in their plans to the green transition, with a total of 340 reforms across the EU. Belgium, Denmark, Finland and Poland devote the largest share of their reforms to green policies. Other Member States, such as Croatia, Cyprus, Hungary and Lithuania, have most reforms related to institutions (including public administration, justice and anti-corruption reforms). While all Member States have included health and social reforms in their plans, Estonia, Sweden and Portugal have allocated the highest share of reforms to this area. Finally, Belgium, Finland, Germany and Slovakia have dedicated a large share of their plans to education and labour market reforms.

[9] See Annex I for more details on the classification.

Figure 3: **Country-level distribution of reforms by key policy domains, in % of total number of reforms**



Source: European Commission services.

3.2 OVERVIEW OF RRF INVESTMENTS BY KEY ECONOMIC SECTORS

As part of the new sectoral database, we also categorised all RRF investment-related measures according to a similar logic as is done with reforms. These include 1,747 investment measures and 212 so-called 'costed reforms'[10], amounting to the RRF total of EUR 655.5 billion.[11]

For the classification of investment measures, we adopted a commodity-based approach, whereby each investment measure is assigned to the sector where the money will actually be spent and where demand is ultimately created. This means that each measure was classified based on the type of goods or services it is expected to purchase or support. For example, if an investment measure finances the building of new schools, the measure will be classified under the 'construction of buildings' sector.[12]

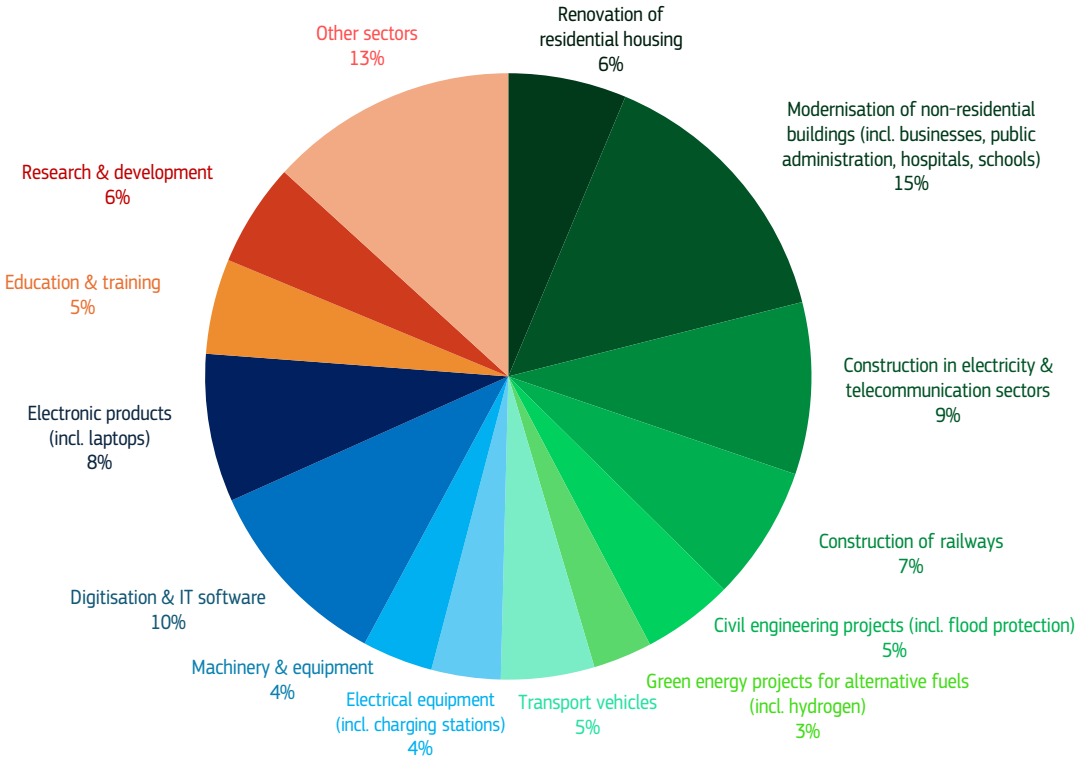
[10] A limited number of reforms (212 out of 1,131) include implementation costs. These 'costed reforms' appear twice in our analysis, as 'reforms' and as 'investments'. Their implementation costs are treated in the same way as investment measures and classified accordingly. Their reform component was classified in line with other reforms and the result is reflected in Figures 2 and 3 above.

[11] The total amount of the RRF stands at EUR 650.4 billion. A small number of Member States have included additional national funds in their plans. This 'over-subscription' amounts to EUR 4.8 billion and is added here.

[12] See Annex I for more details on the classification.

Figure 4 provides an overview of the overall distribution of RRF funds by key sectors, based on an illustrative aggregation into 14 headings[13].

Figure 4: **Overall distribution of investment measures by key sectors, in % of total RRF budget**



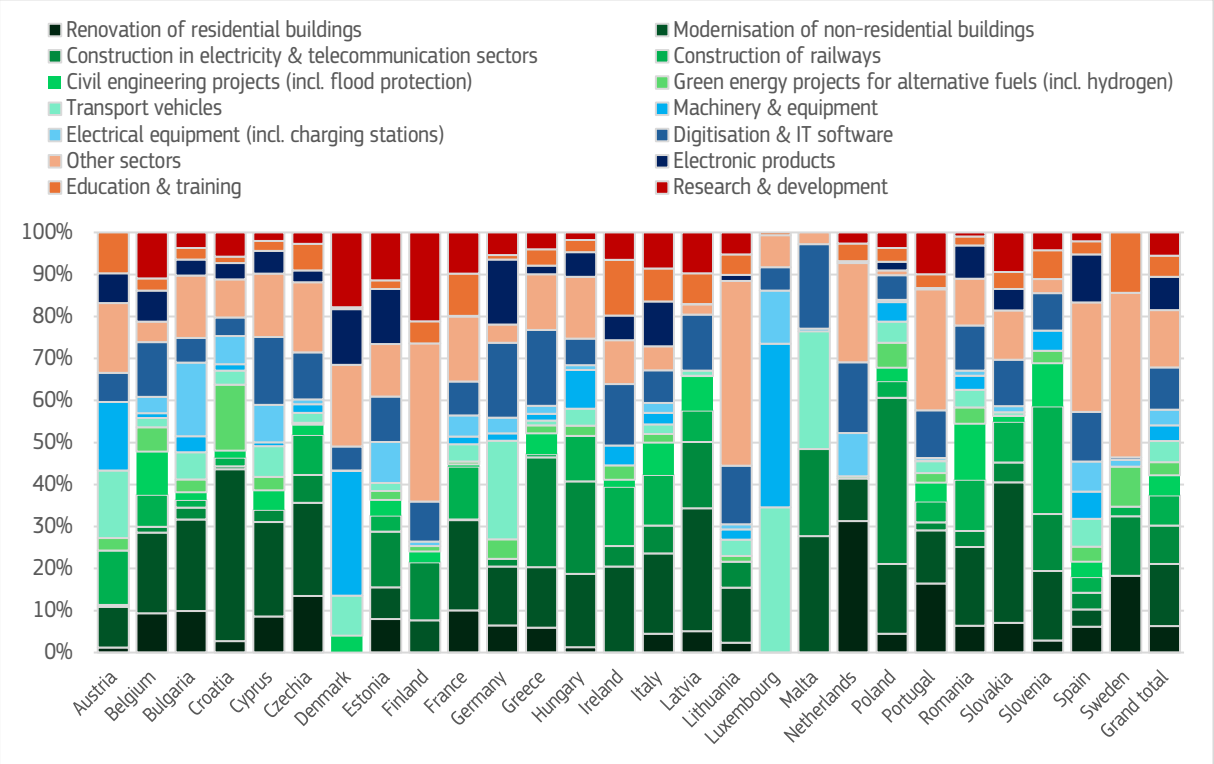
Source: European Commission services.

Looking across the 27 EU Member States, the data shows that the sectors standing to benefit the most from RRF funding include those related to the green transition, such as the energy-efficient modernisation of buildings, construction of electricity and renewable energy projects, railway infrastructure, and investments in sustainable transport vehicles and electrical equipment (e.g. batteries and charging points). Sectors relating to the digital transition also feature prominently, notably IT software, the construction of telecommunication infrastructure and the manufacturing of electronic products and electrical equipment.

The sectoral distribution of RRF funding varies significantly across Member States (see Figure 5 below). Yet, the modernisation of buildings, as well as the renovation and construction of green and digital infrastructure, consistently emerge as the primary investment areas in most countries.

[13] See Annex I for an explanation of the agregation method.

Figure 5: **Sectoral distribution of investment measures by country, in % of national envelope**



Source: European Commission services.

Looking at the specificities of each country, we see that, in Croatia, Latvia, Poland and Slovenia, for example, funding is highly concentrated in one sector, the construction sector, whereas Spain and Austria exhibit a more even allocation of funds across multiple sectors. In the case of Germany, nearly 30 percent of national RRP funding is directed towards the construction and automotive sectors, reflecting the strategic importance of these industries. Moreover, key areas such as digitisation and IT software, electronic products, and research and development feature prominently in Germany's RRP, underscoring their role in fostering technological and industrial transformation.

The above provides a snapshot of what the new RRF sectoral database can show. In the following two Sections 4 and 5, our analysis will focus exclusively on RRF-funded investments, as represented in Figures 4 and 5 above. We thus set aside the impacts of RRF-funded reform measures (as represented in Figures 2 and 3 above), as such analysis goes beyond the scope of this paper. Section 6 will conclude by highlighting potential avenues for future research.

4. MODELLING STRATEGY AND MACRO-ANALYSIS: DIRECT AND SPILLOVER IMPACTS

4.1 THE MODELLING STRATEGY

To assess the macro-economic direct and spillover impacts of RRF-funded investments, we have used the FIDELIO model of the European Commission (Rocchi et al., 2025). FIDELIO is based on official statistics from EUROSTAT's FIGARO global input-output (IO) tables, with a breakdown of 64 NACE industries covering 46 countries (27 Member States, 18 non-EU countries[14] and the rest of the world). Annex II provides a detailed description of the FIDELIO model, Annex III outlines the key assumptions behind the modelling work for this analysis and Annex IV descrives the 64 NACE categories covered in the model.

Designed for policy impact assessments, the model compares counterfactual and baseline equilibriums to assess complex interactions[15] between individual economic agents (households, firms and governments) across national and international markets.[16] The results of the model should not be interpreted as forecasts of future outcomes, but rather as simulation-based assessments of partial potential economic impacts. Moreover, the analysis does not cover the impacts of reforms, nor does it evaluate welfare effects beyond GDP. The estimates presented here are thus likely to underestimate the full impact of the RRF.

In this analysis, the RRF is interpreted as an induced investment shock, which translates into an increase in investment spending and raises aggregate demand. This rise in demand affects both domestic supply and the demand for imported goods. From a geographical perspective, these impacts can be categorised into direct and spillover impacts.

Direct impacts materialise as an increase in production and employment in the industries of the country directly receiving the funds. These effects encompass not only the impact on the funded industry, but also the effect on other related industries, as higher activity generates additional demand for intermediate inputs from domestic supplier industries, thereby stimulating production and employment throughout the national supply chain.

Spillover impacts, in turn, arise from the deep economic integration within the EU, as well as with key non-EU trading partners. The increase in demand will also lead to a higher demand for imported final and intermediate goods. This means that the benefits of the RRF extend beyond national borders, strengthening trade and economic linkages among EU Member States and with non-EU countries.

4.2 MACRO-ANALYSIS: DIRECT AND SPILLOVER IMPACTS

The simulations using the FIDELIO model show that, in 2030, 4 years after the completion of the RRF and 10 years after its launch, the total global cumulative impact of the RRF would amount to EUR 891.7 bn, corresponding to a multiplier of 1.36, out of which EUR 345.5 bn are spillovers. Focusing on the EU alone, the medium-term impact of the RRF would be EUR 685.1 bn, with EUR 138.8 bn in intra-EU spillovers, yielding a multiplier of 1.05.[17]

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[14] United Kingdom, United States, China, Mexico, Canada, Argentina, Brazil, India, South Africa, Saudi Arabia, Indonesia, Australia, South Korea, Japan, Russia, Switzerland, Norway and Türkye.

[15] Besides spillovers, the model also accounts for rebound effects, while quantifying their impacts on employment, economic growth, investments, resource consumption, air emissions and trade balance.

[16] For more details, please see the following link: https://joint-research-centre.ec.europa.eu/scientific-activities-z/trade-and-industrial-policy-analysis/industrial-policy/fidelio-model_en.

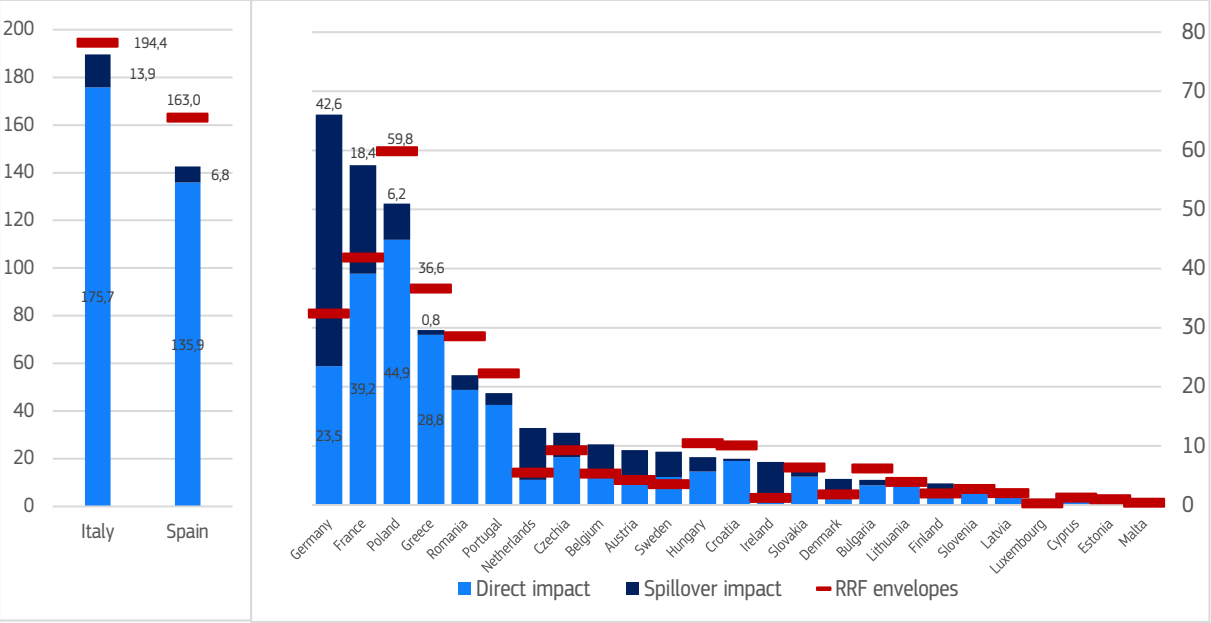
[17] It is worth noting that the highest multiplier effects are observed for Member States classified as "leaders" and "strong" innovators according to the Innovation Scoreboard (2024) published by the Directorate-General for Research and Innovation: https://projects.research-and-innovation.ec.europa.eu/en/statistics/performance-indicators/european-innovation-scoreboard/eis-2024#/eis.

These results are comparable to, yet slightly lower than, those reported under the 'low productivity' scenario of Pfeiffer et al. (2023). The differences can be explained by different modelling approaches. The QUEST model used in Pfeiffer et al. (2023), for instance, applies an aggregate forward-looking modelling framework that includes monetary policy and exchange rate effects. The FIDELIO model used in this paper follows a sectoral modelling approach, with 64 sectors and 46 countries. The data used for the two estimations also differ: this paper analyses the RRF by using precise updated disbursement profiles that reflect the backloading of some measures, similarly to Bańkowski et al. (2024), while Pfeiffer et al. (2023) analyse the *NextGenerationEU* initative, which is larger than the RRF, assuming a uniform distribution of payments over the entire period.[18] Furthermore, both methods rely on different productivity assumptions.[19]

Real GDP in the EU is estimated to be, on average, 0.4 percent higher per year between 2020 and 2030 compared to a scenario without the RRF, with a peak impact of 1.2 percent in 2026. This peak result is higher than the result reported by Bańkowski et al. (2024), but in line with the findings of Pfeiffer et al. (2023). Importantly, sizable growth effects persist over the long term.

Analysing the medium-term effects of the RRF, we find that, given the size of their respective RRPs, Italy and Spain emerge as the two main beneficiaries, with their respective GDP projected to increase by EUR 189.6 bn (8.9% of current GDP) and EUR 142.7 bn (9.5% of current GDP) over the 10-year period. These gains are primarily driven by the direct impacts of their national plans, namely the immediate boost in production and employment in national recipient industries and the increased demand for inputs from domestic supplier industries. Spillover impacts from other RRPs contribute only to a limited extent to the GDP increase for both countries (see Figure 6 below).

Figure 6: **Direct and spillover impacts on value added relative to RRF envelopes, by country (EUR bn)**



Source: European Commission services.

[18] In the mid-term review, however, the assumption of a uniform payment profile is relaxed in favour of a more precise disbursement schedule

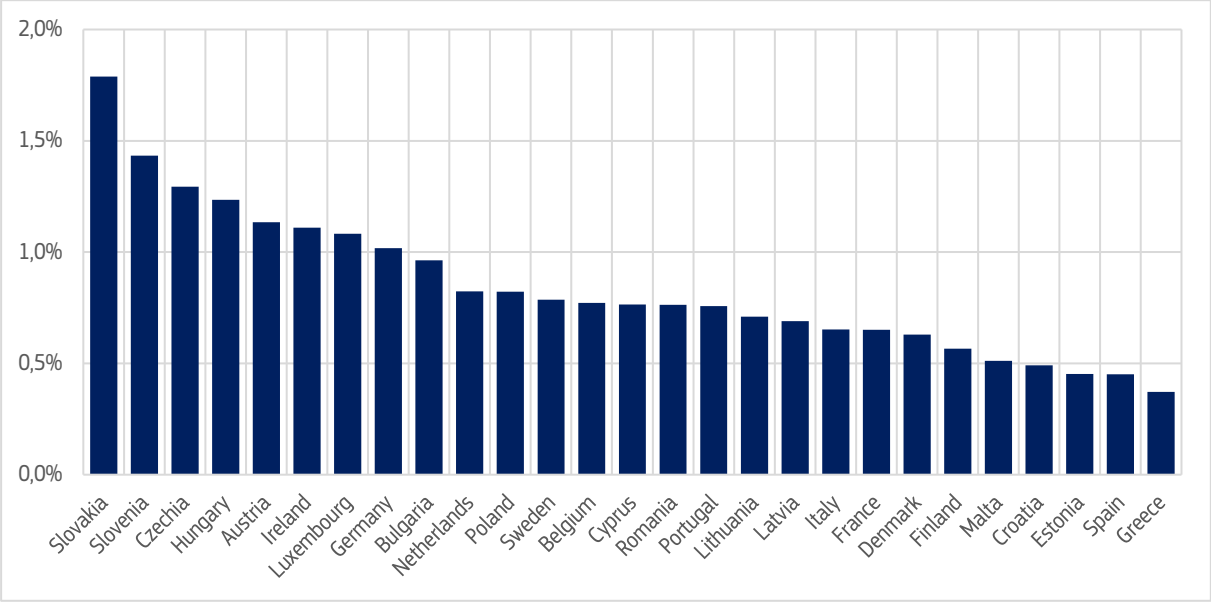
[19] As noted by Pfeiffer et al. (2023), there is no consensus in the literature regarding the productivity effects. To reflect this uncertainty, Pfeiffer et al. present two scenarios (high and low productivity), while this paper uses sector-specific productivity assumptions, with an average elasticity close to the low-productivity scenario of Pfeiffer et al. (2023). Furthermore, the QUEST model treats public capital as a productivity-enhancing, non-rival input (e.g. infrastructure), whereas in the FIDELIO model productivity gains arise from the additional value added generated by RRF funds on R&D investments.

A different picture emerges for the German economy which, due to spillovers from other national plans, becomes the third-largest beneficiary of the RRF. Germany is, in fact, the largest recipient of spillovers across the EU, with an estimated 42.6 billion in spillovers over a 10-year period (see Figure 6), equivalent to 1% of its current GDP (see Figure 7). The total potential macro-economic impact of the RRF in Germany is estimated at EUR 66.1 billion (1.6 % of its current GDP), which is double the size of Germany's RRP.

The ratio of spillovers to direct impact is also important for other Member States that are highly integrated in the Single Market such as the Netherlands (EUR 8.8 bn of spillovers or 0.8% of GDP), Austria (EUR 5.4 bn of spillovers or 1.1% of GDP) and Denmark (EUR 2.4 bn of spillovers or 0.6% of GDP) where spillovers more than double the direct impact. The amount of spillovers is also substantial for France, accounting for 32% of the total impact (EUR 18.4 bn out of EUR 57.5 bn).

The significance of these spillover effects highlights the importance of the Single Market as an amplifier of economic impacts[20]. When considering spillovers relative to GDP (see Figure 7 below), the results show that Slovakia, Slovenia and the Czech Republic are among the EU countries with the highest spillover-to-GDP ratios, at 1.8%, 1.4% and 1.3%, respectively. This can be attributed to their high degree of integration into EU value chains. They are followed by Hungary and Austria, where spillovers amount to 1.2% and 1.1% of their respective GDP.

Figure 7: **Spillovers as a percentage of 2023 GDP**

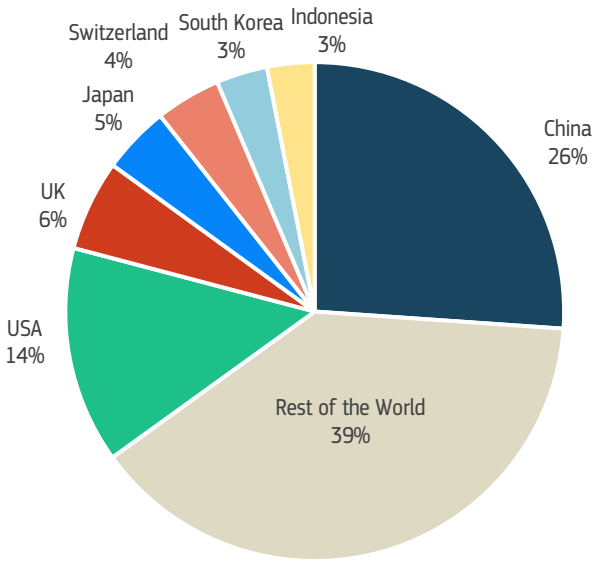


Source: European Commission services.

The vast majority of RRF funds stimulate economic activities and goods produced within the Single Market. Yet, we also observe EUR 206.7 bn of spillovers to non-EU trading partners, an expected outcome given the EU's strong trade integration into global value chains. These include China (26 % of non-EU spillovers), the United States (14 %), the United Kingdom (6%), Japan (5%) and Switzerland (4%) (see Figure 8 below).

[20] See for example Fontagné and Yotov (2025, forthcoming in SMEP): "Assessing the Effectiveness and Potential of the Single Market for New and Existing Members: New Methods and Measures", which shows that on average, the Single Market has led to about 63% increase in trade among EU Member States.

Figure 8: **Non-EU spillovers: share of spillovers going to non-EU countries (total = EUR 206.7 bn)**



Source: European Commission services[21]

China's share in non-EU spillovers also reflects the focus of the RRF on the green and digital transition. Most non-EU spillovers stem from increased demand for inputs in high-tech and medium-high-tech industries, such as the manufacture of computer, electronic and optical products, basic metals, as well as machinery and equipment. This corresponds to the direction of many RRF measures, which drive up demand for products such as solar panels, wind turbines and batteries, for which China currently holds a dominant position in global production and market shares[22]. These findings are thus a further illustration of the EU's external dependencies within sensitive industrial ecosystems[23] (see Arjona et al. (2023)). [24]

[21] The category "rest of the world" includes also Türkiye, Canada, Mexico, Norway, Brazil, Saudi Arabia, Russia, Australia, India, South Africa and Argentina.

[22] See for example Vandermeeren (2024): "Understanding EU-China exposure", which highlighted that China has a very strong position in global production capacities for clean tech products and global market shares in electronic equipment and parts.

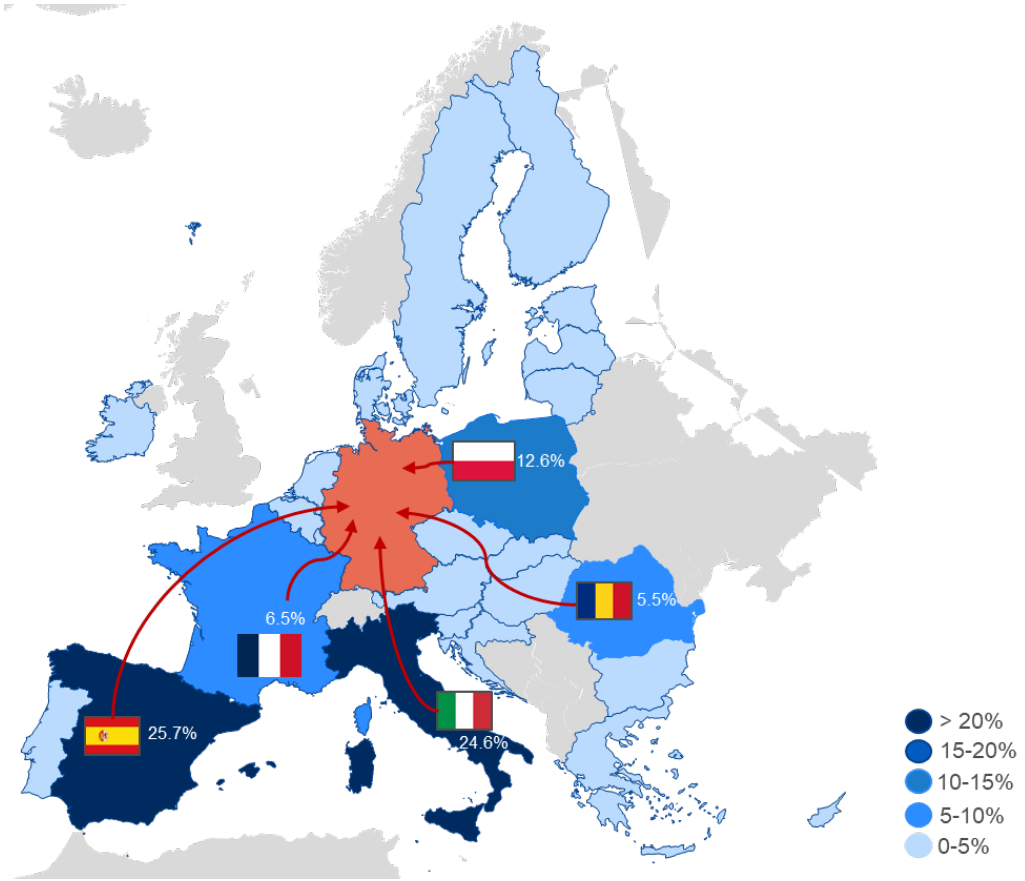
[23] Sensitive industrial ecosystems cover important sectors such as energy-intensive industries, health, as well as those related to the green and digital transition, which are all important beneficiaries under the RRF.

[24] According to the work of Arjona et al. (2023), China is the primary source for approximately 31% of products in sensitive industrial ecosystems. These goods are essential to the EU's twin transition, encompassing both raw materials, such as rare earths, magnesium and manganese, as well as advanced products like solar panels and laptops. Notably, around 80% of EU dependencies on China are subject to 'choke points', where production is concentrated in a few countries that are central nodes in global trade networks.

5. THE CASE OF GERMANY

As mentioned above, the German economy stands out as the largest beneficiary of RRF-induced spillovers, with more than two-thirds of its potential GDP gains originating from the delivery of other countries' RRP. With our work, it is possible to trace the origin of such spillovers, and Figure 9 (see below) shows that the main sources are Spain (25.7% of the total spillovers to Germany), Italy (24.6%), Poland (12.6%), France (6.5%) and Romania (5.5%).

Figure 9: **RRF spillovers to Germany: contributions by country of origin**

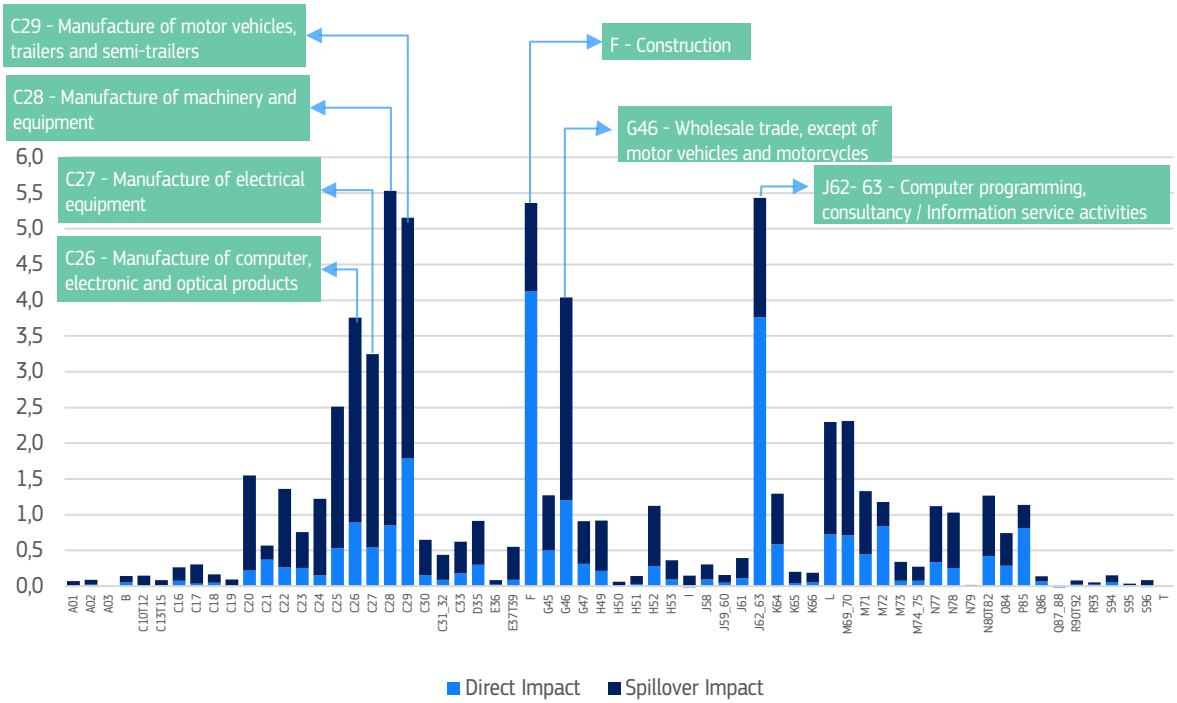


Source: European Commission services.

The significant spillover impacts observed for Germany stem from the deep integration of its industrial sectors with the rest of the EU economy, in particular Spain and Italy. At the core of the German economy lies a globally integrated manufacturing sector, with industries such as automotive, machinery and electrical equipment, that is deeply embedded in complex EU-wide and global value chains. With this work, it is possible to show a sectoral breakdown of the impact of RRF funds, which sheds further light on the central role of Germany as an economic powerhouse and partner (see Figure 10 below).

In Germany, four manufacturing sectors are expected to benefit the most from RRF spillovers from the rest of the EU: the manufacture of machinery and equipment (EUR 4.7 bn), the manufacture of motor vehicles (EUR 3.4 bn), the manufacture of computer, electronic and optical products (EUR 2.9 bn) and the manufacture of electrical equipment (EUR 2.7 bn).

Figure 10: **Main sectors benefitting from the RRF in Germany (EUR bn)**



Source: European Commission services (see Annex IV for the description of the NACE codes).

In addition to significant spillover impacts, the direct impact of the German RRP on these sectors further reinforces their status as important beneficiaries of the RRF. Moreover, the construction sector (housing, buildings, infrastructure) and activities related to computer programming, consultancy, and information services also emerge as significant beneficiaries, with the latter benefiting primarily from Germany's national plan. All these sectors are critical to the German economy, driving both innovation and production in areas essential for the green and digital transition, such as the renovation of buildings, the manufacturing of electric vehicles, and the production of components for IT equipment. Interestingly, many of the sectors benefitting the most from the RRF in Germany are currently undergoing a significant economic transformation[25].

An additional way to exploit the sectoral database is to zoom in on individual sectors in order to estimate more closely RRF deliverables. For instance, looking at the automotive sector (NACE 29.1), the database makes it possible to identify specific measures supporting the uptake of electric vehicles (EVs). Combined with a reading of the measure description, it is then possible to compute the number of EVs financed or subsidised by the RRF overall: more than 1.9 million across the EU. Based on the 2023 market share of German-made EVs within the EU, this corresponds to around 760 000 EVs manufactured in Germany. To give an order of magnitude, this is around 80% of all EVs produced in Germany in the year 2023.[26]

Looking at a related sector, that of charging stations (NACE 27.9), which is critical to support the transition to clean mobility, we find that the RRF is contributing significantly to the deployment of such infrastructure, with over 1.36 million public and private charging stations funded by the RRF across the EU, including 691 500 stations installed in Germany alone. For context, the total number of public charging stations available across the EU stood at 691 000 in the first quarter of 2024.[27]

[25] See Heikkonen, Listl, and Reuter (2025) "Mapping the impact of industrial decline on European regions". DG GROW Economic Briefs.

[26] Based on EUROSTAT data, 966,378 EVs were produced in Germany in 2023 and 608,250 EVs in 2022. Based on EUROSTAT and DESTATIS data, the share of German-made EVs within the EU is equal to 39.5% for 2023.

[27] Source: European Alternative Fuels Observatory.

6. CONCLUSION

This work largely confirms, reinforces and refines previous estimates of macro-economic impacts of the RRF carried out by European Commission services, which provided first orders of magnitude and underlined the importance of spillover effects. The RRF is having a significant impact across the EU and all Member States stand to benefit, to an extent which may go well beyond their national envelopes.

Our simulations indicate that the German economy is the largest beneficiary of spillovers within the EU, with more than two-thirds of its potential medium-term GDP gains driven by the implementation of the RRF in other EU Member States. The German sectors that benefit the most from RRF-funded investments across the EU are the manufacturing sector, including the automotive, machinery, electrical and electronic equipment industries, as well as the construction and IT sectors.

The approach developed here provides new data, findings and techniques that open up new avenues to characterise and visualise the RRF's economic significance. It can pave the way for further, more detailed and sophisticated research, whether for the RRF as a whole, for individual countries or at a sectoral level. Building on this work, several further applications could be envisaged.

First, we could dive deeper into the various Member States and propose more in-depth sectoral analysis. We could also take a thematic approach across countries and focus on specific industries, as was briefly done above for the automotive sector. Such analyses could provide a more detailed understanding of how RRF-funded investments shape sectoral dynamics across economies, offering deeper insights into the spillover effects within the Single Market and the value chain linkages between Member States.

Second, a common challenge in impact assessments has been and remains the evaluation of reforms. By using this sectoral database and linking sectoral data on reforms with macro-economic modelling, new research could offer valuable insights into the impacts of RRF-funded reforms.

Third, a key feature of the RRF is the interplay between reform and investment measures, and the potential benefits arising from their combined implementation. Leveraging the sectoral database, a deeper analysis could be conducted to assess synergies, such as the role of reforms in unlocking investments, with a focus on their combined structural impacts. Looking for instance at the energy sector, it should be possible to estimate how the 126 reforms targeting the energy sector have facilitated the large amount of RRF investments in the sector, and how this interaction has contributed to transform the EU energy market. At a macro level, we could discuss how the RRF has helped not only to preserve the EU Single Market in the aftermath of COVID-19, but also the extent to which it has contributed to its transformation and deepening through the development of (new) value chains.

Fourth, subject to legal and technological feasibility, further research leveraging AI tools could support the development of firm-level databases, compiling data on companies that have received RRF funding, including the amounts disbursed. More than 200,000 documents have already been submitted as part of RRF payment requests. Extracting relevant information from these documents could enable more detailed analysis of the characteristics of beneficiary firms, such as their geographic location and whether they are primarily large exporting champions, small and medium-sized enterprises, or start-ups. Moreover, this could enable the study of the RRF's impacts on key firm-level indicators, including profitability, research and development, and employment. This line of research would complement macro- and sectoral analyses, by offering a more granular view of impacts on the ground.

Fifth, when estimating impacts via the FIDELIO model, this paper assumes full implementation of the 27 national plans, by way of benchmark. The sectoral database can easily be updated in case of changes to the plans, and impact estimates can be adjusted accordingly. Alternative absorption scenarios will be explored in a forthcoming companion paper by the Joint Research Centre.

Sixth, these techniques can serve to document the impacts of the RRF and improve transparency about results, for instance by estimating concrete achievements and deliverables (as was done briefly above for the case of electric vehicles), or by providing further information on beneficiaries at firm level.

Finally, this work benfitted from the swift pace of IT and AI developments. Some further lessons could be drawn for the design of future data collection frameworks related to policy instruments combining reforms and investments, as a way to support evidence-based policy-making.

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ANNEX I: CLASSIFICATION METHODOLOGY

We have developed a new and unique database that classifies all 4,687 measures and submeasures in the national Recovery and Resilience Plans into 241 industries, in line with Eurostat's standard NACE classifications. This classification was performed by using the European Commission's internal AI tools. Specifically, the AI model was prompted with the textual descriptions of the RRF measures contained in the annexes to the Council Implementing Decisions and generated corresponding NACE codes. As the model had been previously trained to recognise NACE classifications, no additional information on the NACE coding system was required and the prompt used was very simple: "Classify the following measure by NACE code providing a primary and a secondary code. Take into account the measure name and the measure description". The AI-generated classifications were then subjected to three rounds of robustness checks by Commission staff to ensure accuracy and consistency. Reform and investment measures were classified separately.

For investments, our classification follows a "commodity" approach rather than an "investment" approach. This means that each measure is categorised according to the sector where it generates demand, rather than the sector that oversees the measure or ultimately benefits from the investment. For example:

- An investment of EUR 10 million under the RRP of Country Y, aimed at developing a web portal for the national administration, is classified under **computer programming, consultancy, and related activities sector** (NACE code 62) rather than **public administration and defence sector** (NACE code 84).

This approach maximises the use of available information by directly linking investments to the sectors where they create demand. By contrast, if we were to classify measures based on the overseeing sector, for instance the 'public administration and defence sector' in this example, we would need to rely on generalised allocation rules for investments of the overseeing sector. These rules estimate how investments (in IT equipment, software, construction, etc.) are typically distributed within this sector, based on historical data. However, this approach often results in less precise information about which parts of the economy are directly impacted. By instead classifying each measure according to the specific goods or services it finances, we can more accurately identify the sectors where demand will increase. This method provides a clearer and more targeted picture of how RRF investments affect different areas of the economy, ensuring a more reliable assessment of their economic impact.

An additional challenge in classifying the measures arises when a single measure is linked to multiple sectors. To address this, we have created two classification categories: primary NACE code and secondary NACE code. The most significant expenditure within a measure is assigned as the primary NACE code, while smaller expenditures are assigned as secondary NACE codes. For example:

If country X's RRP invests EUR 200 million in Measure M, which includes €150 million for constructing a new railway network and EUR 50 million for acquiring two new trains, the classification must reflect these distinct components. In this case:

- The construction of the railway network would be classified under **construction of railways** (NACE code 42.12).
- The acquisition of new trains would be classified under **manufacture of railway locomotives and rolling stock** (NACE code 30.2).

The classification of investments was limited to a maximum of two sectors since very few investment measures generate demand across more than two sectors. A majority (54%) is found to impact only one sector.

For reforms, our classification follows a similar but different approach. Each reform (sub)measure is categorised based on the sector targeted by the reform. For example:

- If the objective of reform T in the RRP of country X is the reduction of bureaucracy by eliminating double and multiple reporting, we classify this measure under **regulation and contribution to more efficient operation of business** (NACE code 84.13).
- Similarly, if the aim of reform J is to increase the collection rates for beverage packaging, we will classify this measure under **recovery of sorted materials** (NACE code 38.32).

If a reform measure is linked to multiple sectors, the primary NACE code is assigned to the sector primarily targeted by the reform. Sectors more indirectly targeted are assigned to the secondary NACE code category. For example:

If the objective of reform E is the expansion of renewable energy in electricity supply this measure could be classified under either:

- **Electricity power generation, transmission and distribution** (NACE code 35.1), or
- **Manufacture of machinery and equipment** (NACE code 28), as it will also impact the production of solar panels and wind turbines.

In such cases, we allocate the **primary NACE code** to the industry which is the primary target of the reform (**Electricity power generation, transmission and distribution)** and the **secondary NACE code** to the industry which is more indirectly targeted by the reform **Manufacture of machinery and equipment** (NACE code 28).

The classification of reforms has also been limited to a maximum of two sectors per measure, since no reform measure was found to target more than two economic sectors. A majority of measures (63%) is found to target only one economic sector.

To ensure clarity and analytical focus, the description of the database above, as well as Figures 2, 3, 4 and 5, rely exclusively on the primary NACE codes. Likewise, the subsequent analysis only uses the primary NACE code category for now. Secondary NACE codes will be considered, for example, in a companion paper by the JRC, which will include robustness checks based on alternative weighting schemes for primary and secondary classifications in the estimations.

For visualisation purposes, some NACE categories have been regrouped in Figures 2 and 3, and separately in Figures 4 and 5. The details of these groupings are provided below.

Overall distribution of reforms by key policy domains (Figures 2 and 3)

Broader policy domains (Figure 3)	Policy domains (Figure 2)	NACE codes
Green	Energy	35
	Transport	29, 49, 50, 51, 52
	Modernisation of buildings	41, 42, 43
	Water supply & waste	36, 38, 39
	Agriculture	1, 2, 3
	Other (green)	5, 6, 9, 91.04
Business	Digital & IT sector	61, 62, 63
	Simplification of regulation (cross-cutting)	84 (sub-categories)
	Research & Development	72, 77
	Professional and technical activities	69, 70, 71, 74
	Financial markets	64, 65, 66
	Other (business)	10, 16, 20, 21, 25, 26, 28, 47, 55, 79
Institutions	Public administration	102
	Justice, anti-corruption and anti-fraud	84 (sub-categories)
	Taxation	84 (sub-categories)
	Public spending reviews	84 (sub-categories)
	Implementation of the plan	84 (sub-categories)
	Public procurement	84 (sub-categories)
	Other (institutions)	84.21
Health & social	Health	86
	Social security (e.g. pensions)	84.3
	Social (incl. long term and child care)	87, 88
	Culture	59, 60, 90, 91, 93
	Social Housing	68
	Other (health & social)	2, 84.2, 84.25
Skills & labour	Education	85
	Labour market	84 (sub-categories)

Note: NACE code 84 was split based on the CID description to match the categories above.

Overall distribution of investment measures by key sectors (Figures 4 and 5)

Key sectors	NACE codes
Renovation of buildings	41, 43*
Construction in electricity & telecommunications sectors	42.22
Construction of railways	42.12
Civil engineering projects (incl. flood protection)	42.1, 42.11, 42.13, 42.9, 42.91, 42.99
Green energy projects for alternative fuels	42.21
Transport vehicles	29, 30
Electrical equipment (incl. charging stations)	27
Machinery & equipment	28
Digitalisaion & IT software	62, 63
Electronic products (incl. laptops)	26
Education & training	85
Research & development	72

Note: Renovation of buildings (codes: 41, 43) were further split into renovation of residential housing and non-residential buildings (incl. businesses, public administration, hospitals, schools) based on the CID description of each measure.

ANNEX II: THE FIDELIO MODEL

The FIDELIO model is a multi-country, multi-sector dynamic econometric general equilibrium model, developed by the Joint Research Centre of the European Commission for policy impact assessment, providing industrial-, country-, and time-specific simulations (Rocchi et al., 2025, Rocchi et al., 2019).

The model is based on official Eurostat's statistics from the FIGARO database, covering 64 industries (NACE Rev. 2) and 45 countries (EU-27 Member States, its main 18 non-EU trading partners, and an aggregate region representing the rest of the world (Remond-Tiedrez and Rueda-Cantuche, 2019)). The and other data come from Eurostat and the OECD official statistics.

The FIDELIO model has been used to support policy analysis related to topics such as the economic impact of state aid on the green transition and the effects on the EU Single Market (Rueda-Cantuche, et al., 2024), for calculating the rebound effects of food waste prevention (Albizzati et al., 2022), to analyse the employment and value added impacts of the stricter energy efficiency requirements for dishwashers, washing machines, and washer dryers (Rocchi et al., 2019), to evaluate the potential macro-economic benefits of eliminating the regulatory restrictiveness in business services in the European countries (Barbero et al., 2022), and to estimate the macro-economic impact of US trade tariff increases on steel and aluminium (Salotti et al., 2019).

Modular framework

FIDELIO is primarily a demand-driven[28] input-output based model, which is structured into four modules that can be activated and deactivated as desired depending on the case study. This modular structure enhances the ability to decompose model outcomes, with greater or lesser degree of sophistication, thus improving transparency and facilitating the interpretation of simultaneous economic impacts within a CGE framework across industries and countries, as described below.

Module 1: Standard input-output module (direct and spillover impacts)

This module is the core of the FIDELIO model and it is used for standard input-output analysis. It applies the Leontief model, which assumes exogenous components of final use and production technology. The model captures inter-industry spillovers impacts from intermediate consumption and allows for the tracking of economic interdependencies among industries, both domestically and across different countries. The results of this module focus on the direct and spillover impacts of changes in the final use of products.

Module 2: Investment input-output module (investment multiplier impacts)

The investment input-output module goes beyond the basic input-output analysis by considering the impacts of investments. This module endogenises investors and activates capital accumulation. This allows analysts to separate investment impacts from inter-industry spillovers impacts, making it particularly relevant for policy studies that aim to influence investment decisions.

Module 3: National accounting matrix module (distributional impacts)

This module includes the data displayed in a national accounting matrix (also sometimes known as a social accounting matrix), which endogenises final use (households and governments) and income distribution. This module captures the behaviour of all economic agents, enabling the model to calculate the income distributional impacts of EU policies. It enriches the range of analyses that FIDELIO can support by addressing redistributive impacts.

[28] The model can also model supply shocks, such as the evaluation of the impacts of state aid in specific industries (Rueda-Cantuche et al., 2024) or the impacts derived from green and digital transitions (e.g. greener technologies).

Module 4: Full econometric and price module (substitution and rebound impacts)

The full model includes all econometric components, the dynamic[29] nature of the model and endogenous prices. It introduces substitution and rebound impacts between different factor inputs, goods, and between imported and domestically produced goods. Although this module provides the most realistic scenarios, it involves complex interactions and makes the results more challenging to interpret. Nonetheless, the dissociation of price effects from other effects is possible by comparing it with previous modules, a unique feature of the FIDELIO model. This is the module used in the analysis of the economic impacts of RRF funds.

The four modules can be activated or deactivated, enabling a gradual transition from a standard multiregional input-output model[30] to a full econometric input-output model[31], passing through a CGE models. [32] This approach effectively combines econometric techniques with two widely used impact assessment models, i.e. input-output models (IO) and computable general equilibrium (CGE) models, which extend IO analysis by incorporating additional economic agents (e.g. institutional sectors: households, government, etc.), secondary income accounts and price adjustment mechanisms in the short and long run via econometric methods (Koks et al., 2016).

Model structure

FIDELIO is structured in four different blocks: production (firms); trade; government, households and prices.

Production block

The production block is defined by firms that adjust their production to meet the total demand for domestic products. This demand includes several components: intermediate inputs, investment, changes in inventories, household consumption, government consumption, and foreign demand. Firms use production functions with nested constant elasticities of substitution (CES) to model substitution among different inputs, such as energy and materials. This block is crucial for understanding how firms optimise production and respond to both domestic and international market conditions.

Trade

The trade block endogenises exports, which are considered as the mirror flow of imports as shown by the FIGARO database. Bilateral export transactions are calculated based on bilateral imports through a two-step process. First, the model determines the share of imports in the total demand for each commodity. It then allocates imports to the country of origin. Armington elasticities guide import substitution between domestic and imported goods, considering differences due to consumer preferences or historical positions. This aspect allows FIDELIO to analyse trade dependencies and flows between countries.

[29] The dynamic nature of the FIDELIO model solves recursively baseline and simulation runs over time. The model's dynamics is partially driven by the evolution of variables such as population growth and capital accumulation. Furthermore, the FIDELIO model integrates neo-Keynesian principles to enhance its representation of economic behaviour.

[30] A standard multi-country input-output model captures interdependencies among different countries and industries by tracking the flow of goods and services across economies. The model is static and it is based on national accounts data and international trade statistics. Prices do not react to supply and demand shocks and therefore, IO models tend to be used for short-run impacts.

[31] The integration of econometric techniques in input-output models and CGE models allow them to have dynamic behavioural responses, price adjustment mechanisms and structural changes over time, making the models more flexible in capturing economic fluctuations and policy impacts.

[32] Computable general equilibrium (CGE) models assume that agents (households, firms, governments) make decisions based on optimising behaviour, leading to more complex interactions. CGE models make price formation a central aspect of the analysis while input-output models do not.

Government

In the government block, the government generates revenue from multiple sources: operating surplus, taxes on production, taxes on products, social security contributions, and household income taxes. These revenues finance government expenditure, which includes interest, capital formation, transfers to households, and government consumption. The model calculates the budget balance, measured by the difference between revenue and expenditure, influencing public debt levels. By capturing governmental activities, this block describes how policy decisions affect and are influenced by fiscal health.

Households

Households receive income through wages, operating surplus shares, property income, and transfers (governmental and non-governmental), after deducting taxes and social security contributions. Disposable income is then divided between consumption and savings, with the model considering factors such as unemployment rates, interest rates, and debt-to-GDP ratios that affect the households' propensity to consume or save.

Once the total level of household consumption is determined, the FIDELIO model describes how this expenditure is optimally allocated across different commodities to maximise utility under an income constraint. The allocation of consumption across commodities can be implemented using two[33] approaches, i.e. either a linear expenditure system (LES) or an almost ideal demand system (AIDS).

Prices

The price block in the FIDELIO model serves to incorporate general equilibrium effects into the system by representing a comprehensive set of prices, divided into two categories: prices constructed through accounting equations (accountancy prices) and behavioural prices (production prices, interest rates and wages).

Accountancy prices are derived from accounting relationships and include the consumer price index (HCP), which is a weighted average of commodity prices. Besides, the purchaser's price of a commodity is calculated as a mix of the basic price, transportation, trade margins, and taxes minus subsidies. The accountancy prices are derived from straightforward accounting identities, not influenced by economic behaviours.

In contrast, behavioural prices, such as the basic price, are derived from rational economic behaviour and are crucial to determine, for instance, one of the components of the purchaser's prices (accountancy price). There are three main behavioural prices in FIDELIO: production prices, interest rates and wages.

- Production prices are set by producers aiming to maximise profits in an oligopolistic market structure (*à la Cournot*). They consider competitors' prices and assume a negative relationship between price and demand, determining optimum prices as a mark-up over marginal costs.
- Interest rates are modelled following the Taylor rule, which emphasises the role of the central bank in setting interest rates based on inflation and unemployment rather than simply aligning savings and investment. This reflects a more realistic depiction of modern economies, where central banks have a significant impact on the supply of credit beyond immediate savings, thereby maintaining monetary equilibrium. The cost of capital is defined similarly across industries, using a standard specification of the user cost of capital, which includes depreciation and interest rates.
- Wages can be represented using either the Phillips curve or the Wage Setting (WS) curve, each impacting the equilibrium unemployment rate differently. The model allows for nominal wages to depend on expected consumption prices, labour productivity, and unemployment rates, as per a chosen curve specification. The Phillips curve does not necessitate unit wage indexation on prices and productivity, contrary to the WS curve requirements for long-term steady states.

[33] Under a LES, household consumption of a commodity is modelled by taking into account a fixed minimum purchase level for necessities, distinguishing between essential and luxury goods. Under an AIDS model (Mongelli et al., 2010), FIDELIO offers more flexibility by allowing consumption shares to be influenced by both relative prices and income levels. The AIDS model calculates consumption shares as a function of price and income, thus allowing for a wider range of consumer behaviours. This model is econometrically estimated.

ANNEX III: MAIN MODELLING ASSUMPTIONS

In our analysis, we simulated the impact of a demand-driven investment[34] shock of EUR 655.5 billion, comprising EUR 364.5 billion in non-repayable grants[35] and EUR 291 billion in loans, financed through the RRF over the 2020-2026 period. Since loans and grants may exhibit distinct macro-economic impacts, they have been modelled differently, as suggested by Bankowski et al. (2022) and Canova and Pappa (2025). Grants are treated as non-debt-creating transfers for recipient Member States, whereas loans are modelled as repayable over a period of up to 30 years at a fixed annual interest rate of 3.15 %, in line with the ECB's main refinancing operations rate.

For analytical purposes, and in line with Pfeiffer et al. (2023), and Barbero et al. (2024), we assume that RRF grants are financed through long-term EU debt. The repayment of this debt is covered by Member States' contribution to the EU budget, which are, in turn, funded via lump sum taxes. Repayments are scheduled to begin at the end of the current multiannual financial framework in 2027 and and will continue on a linear path until 2058. Under this approach, households bear the cost of the grants, which reduces their disposable income and negatively affects their consumption and savings behaviour, partially offsetting the RRF positive economic effects. In contrast, loans, while also injecting funds into the economy, are modelled to reflect their impact on public debt. Since they require repayment, loans raise future fiscal obligations, potentially creating budgetary constraints that may limit their long-term economic stimulus effect to a greater extent than grants.[36]

By introducing RRF funds as an exogenous investment shock, the model allows us to trace the transmission of economic impacts across sectors and countries, including productivity gains. These extra gains are derived from those RRF funds that allow developing and/or upgrading new technologies, products, and services, thereby contributing to long term economic growth across the EU (Pfeiffer et al. (2023), Bankowski et al. (2024). In FIDELIO, productivity gains come from the extra value added generated by the new RRF funds on R&D investments. This effect is econometrically estimated using detailed information by country and industry on the R&D investments and their subsequent extra value added impacts. [37]

Finally, we assume full implementation of the 27 national plans, as of mid-January 2025. The database already accounts for past amendments and updates of plans prior to that date and it can be updated to reflect possible future changes to plans. A forthcoming companion paper by the Joint Research Centre will discuss alternative absorption scenarios.

[34] According to the system of national accounts, investment grants consist of capital transfers made by governments or international organisations to other institutional units to finance all or part of the costs of their acquiring non-financial assets (par. 11.247, 2025 System of National Accounts, UN).

[35] The RRF includes EUR 359 billion in grants, as well as EUR 5.5 billion in nationally financed oversubscription, which is treated as grant-equivalent for estimation purposes.

[36] The economic impact of RRF grants and loans is expected to differ significantly, not only because of their distinct financing structures, but also due to the different purposes they serve. The FIDELIO simulation framework captures these differences by using detailed expenditure data disaggregated by sector for both grants and loans. This granularity allows for a more refined and accurate assessment of their respective impacts, avoiding reliance on broad or simplifying assumptions.

[37] The productivity effects are based on the estimates by Pedauga et al. (2025), who, similarly to Moncada-Paternò-Castello et al. (2020), use historical data from the Community Innovation Survey to estimate returns on R&D investments. Expected sectoral R&D intensity is derived from the GERD database, while the econometric methodology follows established approaches, including those of Papke and Wooldridge (2008) and Delgado et al. (2018). The resulting sector-specific elasticities vary significantly, with for example, higher values for high-tech sectors (0.161) and lower ones for the construction sectors (0.0083). When combined with the actual sectoral distribution of RRF investments, the average elasticity is 0.028 which is close to the low productivity scenario elasticity of Pfeiffer et al. (2023).

ANNEX IV: FIDELIO NACE CODES

For the economic analysis based on the FIDELIO model, we have merged the 241 NACE categories into the following 64 categories:

NACE A*64	NACE A*64 Name
A01	Crop and animal production, hunting and related service activities
A02	Forestry and logging
A03	Fishing and aquaculture
B	Mining and quarrying
C10T12	Manufacture of food products, beverages and tobacco products
C13T15	Manufacture of textiles, wearing apparel and leather products
C16	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials
C17	Manufacture of paper and paper products
C18	Printing and reproduction of recorded media
C19	Manufacture of coke and refined petroleum products
C20	Manufacture of chemicals and chemical products
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations
C22	Manufacture of rubber and plastic products
C23	Manufacture of other non-metallic mineral products
C24	Manufacture of basic metals
C25	Manufacture of fabricated metal products, except machinery and equipment
C26	Manufacture of computer, electronic and optical products
C27	Manufacture of electrical equipment
C28	Manufacture of machinery and equipment n.e.c
C29	Manufacture of motor vehicles, trailers and semi-trailers
C30	Manufacture of other transport equipment
C31_32	Manufacture of furniture; other manufacturing
C33	Repair and installation of machinery and equipment
D35	Electricity, gas, steam and air conditioning supply
E36	Water collection, treatment and supply
E37T39	Sewerage; waste collection, treatment and disposal activities; materials recovery; remediation activities and other waste management services
F	Construction
G45	Wholesale and retail trade and repair of motor vehicles and motorcycles
G46	Wholesale trade, except of motor vehicles and motorcycles
G47	Retail trade, except of motor vehicles and motorcycles

H49	Land transport and transport via pipelines
H50	Water transport
H51	Air transport
H52	Warehousing and support activities for transportation
H53	Postal and courier activities
I	Accommodation; food and beverage service activities
J58	Publishing activities
J59_60	Motion picture, video and television programme production, sound recording and music publishing activities; programming and broadcasting
J61	Telecommunications
J62_63	Computer programming, consultancy and related activities; information service activities
K64	Financial service activities, except insurance and pension funding
K65	Insurance, reinsurance and pension funding, except compulsory social security
K66	Activities auxiliary to financial services and insurance activities
L	Real estate activities
M69_70	Legal and accounting activities; activities of head offices; management consultancy activities
M71	Architectural and engineering activities; technical testing and analysis
M72	Scientific research and development
M73	Advertising and market research
M74_75	Other professional, scientific and technical activities; veterinary activities
N77	Rental and leasing activities
N78	Employment activities
N79	Travel agency, tour operator reservation service and related activities
N80T82	Security and investigation activities; services to buildings and landscape activities; office administrative, office support and other business support activities
O84	Public administration and defence; compulsory social security
P85	Education
Q86	Human health activities
Q87_88	Social work activities
R90T92	Creative, arts and entertainment activities; libraries, archives, museums and other cultural activities; gambling and betting activities
R93	Sports activities and amusement and recreation activities
S94	Activities of membership organisations
S95	Repair of computers and personal and household goods
S96	Other personal service activities
T	Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use
U	Activities of extraterritorial organisations and bodies

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